Supplemental Tables and Figures:

Predicting hepatotoxicity using ToxCast *in vitro* bioactivity and chemical structure

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Keywords: machine learning, bioactivity, hybrid descriptors, QSAR, hepatotoxicity.

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Table S1. The liver effects and curated corresponding lesion categories. The table lists 145 liver histopathology endpoints observed in rats after chronic oral administration of the test chemicals in ToxRefDB and their corresponding three broad categories, including hypertrophy, injury and proliferative lesions, based on the terminology defined by domain experts.

Table S2. The maximum predictive performance of different classification methods for the balanced data. The table summarizes the results of 10-fold cross-validation testing for the three imbalanced data sets (Toxicity) including: hypertrophy (Hyp), injury (Inj), and proliferative lesions (Pro). The results are broken down by different classification algorithms (Classifier) including: linear discriminant analysis (LDA), support vector classifier with linear kernel (SVCL0), support vector classifier with radial basis function kernel (SVCR0), Naïve Bayes (NB), classification and regression trees (CART0), k-nearest neighbor classification (KNN1) and an ensemble of all classifiers (ENSMB). The results for each data set and classification method are summarized by the mean balanced accuracy (BA), the number of descriptors that produced the best BA, mean sensitivity and mean specificity. The standard deviation is given in parentheses. The performance results for three types of descriptors are shown: chemical structure descriptors (CHM), ToxCast bioactivity descriptors (BIO), and hybrid chemical and bioactivity descriptors (BC). For instance, the first row shows the 10-fold cross-validation testing performance for predicting rat liver hypertrophy by CART0 using the imbalanced data sets. The maximum BA of CART0 for predicting hypertrophy was 0.81, 0.79, 0.80 using 45 hybrid descriptors, 60 bioactivity descriptors, and 55 chemical

descriptors, respectively. Furthermore, the mean sensitivity/specificity for the hybrid, bioactivity and chemical by CART0 was 0.80/0.84, 0.73/0.89, and 0.81/0.80, respectively.

Table S3. 36 bioactivity descriptors most frequently used in classifiers. The table lists the detailed information for the 36 bioactivity descriptors, including the cluster group of the heatmap (Figure 4), assay technology platform, assay target gene, assay target family, and the cell line was used for the assay.

Table S4. 55 chemical descriptors most frequently used in classifiers. The table lists the detailed information for the 55 chemical descriptors, including the cluster group of the heatmap (Figure 5), descriptor source, descriptor SMARTS string and graph for the general structure.

Table S5. The maximum predictive performance of different classification methods for the imbalanced ToxCast Phase I data. The table summarizes the results of 10-fold cross-validation testing for the three imbalanced data sets (Toxicity) including: hypertrophy (Hyp), injury (Inj), and proliferative lesions (Pro). The results are broken down by different classification algorithms (Classifier) including: linear discriminant analysis (LDA), support vector classifier with linear kernel (SVCL0), support vector classifier with radial basis function kernel (SVCR0), Naïve Bayes (NB), classification and regression trees (CART0), k-nearest neighbor (KNN1) and an ensemble of all

classifiers (ENSMB). The results for each data set and classification method are summarized by the mean balanced accuracy (BA), the number of descriptors that produced the best BA, mean sensitivity and mean specificity. The standard deviation is given in parentheses. The performance results for three types of descriptors are shown: chemical structure descriptors (CHM), ToxCast bioactivity descriptors (BIO), and hybrid chemical and bioactivity descriptors (BC).

Figure S1. Cross-validation performance results using ToxCast Phase I data only. The figure shows the 10-fold cross-validation performance results for the three data sets (rows) including: hypertrophy (Hyp), injury (Inj), and proliferative lesions (Pro), and three types of descriptors (columns) including: chemical structure descriptors (chm), ToxCast bioactivity descriptors (bio), and hybrid chemical and bioactivity descriptors (bc). The x-axis shows the number of descriptors (N_d) and the y-axis shows the balanced accuracy (Bal Acc). Each curve shows the relationship between the balanced accuracy and number of descriptors for different classification algorithms including: linear discriminant analysis (LDA), support vector classifier with linear kernel (SVCL0), support vector classifier with radial basis function kernel (SVCR0), Naïve Bayes (NB), classification and regression trees (CART0), k-nearest neighbor classification (KNN1) and an ensemble of all classifiers (ENSMB). The number of observations (N_{obs}) used for each data set are shown in the title.

Table S1.

Effect	Category
Absolute	Hypertrophy
Accentuated Lobular Pattern	Injury
Adenocarcinoma	Proliferative lesion
Adenoma	Proliferative lesion
Adenoma/Carcinoma Combined	Proliferative lesion
Angiectasis	Injury
Apoptosis	Injury
areas of collapse	Injury
Arteritis	Injury
Atrophy	Injury
Atypia Cellular	Proliferative lesion
Atypia Nuclear	Proliferative lesion
Atypical Focus	Proliferative lesion
Autolysis	Injury
Basophilic Focus	Proliferative lesion
Benign liver cell tumors	Proliferative lesion
Carcinoma	Proliferative lesion
Carcinoma NOS	Proliferative lesion
Cellular Alteration	Proliferative lesion
Cholangiocarcinoma	Proliferative lesion
Cholangioma	Proliferative lesion
Clear Cell Focus	Proliferative lesion
Collapsed interlobular stroma; 1 male; In area of necrosis	Injury
Congestion	Injury
Cytologic Alterations	Injury
Cytomegaly	Hypertrophy
Cytoplasmic Alteration	Proliferative lesion
Deformity	Injury
Degeneration	Injury
Depletion Glycogen	Injury
Diffuse Hepatocellular swelling	Injury
Dilatation	Injury
Disassociation of liver cells	Injury
Discolored	Injury
Dissociation of hepatocellular cords	Injury
dissociation of liver cell plates	Injury
Distended	Hypertrophy
Dysplasia	Injury
Ectasia	Injury

Edema	Injury
enzyme induction	Hypertrophy
Eosinophilic Focus	Proliferative lesion
Erythrophagocytosis	Injury
Foam Cell Aggregation	Injury
Focal acidophilic	Proliferative lesion
Focal Cellular Change	Proliferative lesion
Focal coagulation	Injury
Giant Cell	Injury
Glycogen Deposition	Injury
Glycogen Liberation	Injury
granular amphophilic cytoplasm	Injury
Granulation	Injury
Granules - Terminal sacrifice	Injury
Granuloma	Injury
Granulomatous Hepatitis	Injury
Ground Glass appearance	Injury
Hemangioma	Proliferative lesion
Hemangiosarcoma	Proliferative lesion
Hematopoietic Cell Proliferation	Proliferative lesion
Hematopoietic Cell Proliferation Erythrocytic	Injury
Hematopoietic Cell Proliferation Granulocytic	Proliferative lesion
Hemorrhage	Injury
Hemosiderin Laden Macrophages	Injury
Hemosiderosis	Injury
hepatic disfunction, considered secondary to hepatocellular injury	Injury
hepatic dysfunction, secondary to hepatocellular injury	Injury
Hepatitis	Injury
Hepatoblastoma	Proliferative lesion
Hepatocellular Adenoma	Proliferative lesion
Hepatocellular Carcinoma	Proliferative lesion
Hepatocholangiocarcinoma	Proliferative lesion
Hepatocholangioma	Proliferative lesion
Hepatocyte, Multinucleate	Proliferative lesion
hepatocytolysis	Injury
Hepatodiaphragmatic Nodule	Injury
Hepatoma	Proliferative lesion
hepatoma, NOS	Proliferative lesion
Hepatomas	Proliferative lesion
Histiocytic Sarcoma	Proliferative lesion
Histiocytosis	Injury
Hyalinization	Injury

Hyperplasia	Injury
Hyperplastic Nodule (Pre-Neoplastic)	Proliferative lesion
Hypertrophy	Hypertrophy
Hypoplasia	Injury
Inclusion Body Intracytoplasmic	Injury
increase in number of mice with one or more primary tumors	Proliferative lesion
increased Kupffer cell nodules	Injury
increased ploidy	Hypertrophy
Infiltration Cellular	Injury
Inflammation	Injury
intrahepatic bile duct oval cells	Proliferative lesion
Karyomegaly	Hypertrophy
Leukemia Lymphocytic	Proliferative lesion
Leukemia Mononuclear	Proliferative lesion
Leukocytosis	Injury
Leukostasis, could reflect nonspecific hepatocyte injury	Injury
Lipofuscin Deposits	Injury
Lymphoid Aggregation	Injury
Macrophage Accumulation	Injury
Macrophages	Injury
Malignant liver cell tumors	Proliferative lesion
Margination	Injury
Mass	Injury
Mass(es)/raised areas/swelling/ nodular areas	Injury
Megalocytosis	Hypertrophy
Metaplasia	Injury
minimal microscopic changes in liver (gender not specified)	Injury
Mitosis	Proliferative lesion
Mitotic Alteration	Proliferative lesion
Mixed Cell Focus	Proliferative lesion
Mixed Tumor Malignant	Proliferative lesion
Mixed Tumor NOS	Proliferative lesion
Multinucleated	Proliferative lesion
multple cell foci of the hepatocytes	Proliferative lesion
myelopoiesis	Injury
Necrobiosis	Injury
Neoplasm NOS	Proliferative lesion
Neoplastic nodule	Proliferative lesion
Nodules and Carcinoma combined	Proliferative lesion
Nuclear Alteration	Injury
Parenchymal tumors (benign & malignant)	Proliferative lesion
Pigmentation	Injury
Pleiomorphism	Proliferative lesion

Plug	Injury
Porphyrins	Injury
Portal Tract Changes	Injury
Proliferation	Proliferative lesion
pyknotic or karyohectic nuclei with condensed cytoplasm	Injury
Rarefaction	Injury
Regeneration	Proliferative lesion
Relative to Body Weight	Hypertrophy
Relative to Brain Weight	Hypertrophy
Sarcoma	Proliferative lesion
Sinusoidal Space	Injury
Syncytial Alteration	Proliferative lesion
Telangiectasis	Injury
Thickened	Injury
Thrombosis	Injury
Tissue alteration	Injury
Toxic Hepatopathy	Injury
Tubular collapse; 1 dog/sex	Injury
Type A nodule, significant trend only	Proliferative lesion
Vacuolization Cytoplasmic	Injury
Vascular macrophages	Injury

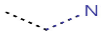
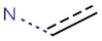
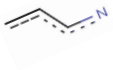
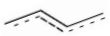
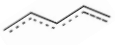
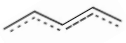
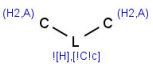
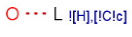
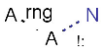
Table S2.

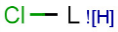
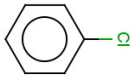
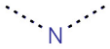
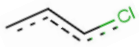
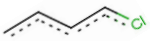
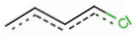
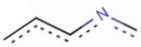
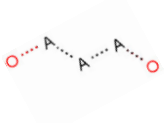
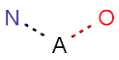
		# desc.			BA			Sensitivity			Specificity		
		BIO	**CHM**	**BC**	**BIO**	**CHM**	**BC**	**BIO**	**CHM**	**BC**	**BIO**	**CHM**	**BC**
tox	**cl**												
Hyp	CART0	60	55	45	0.79 (0.08)	**0.80 (0.09)**	**0.81 (0.11)**	0.73 (0.13)	0.81 (0.16)	0.80 (0.18)	0.89 (0.11)	0.80 (0.17)	0.84 (0.13)
	ENSMB	60	60	60	**0.79 (0.07)**	0.79 (0.09)	**0.81 (0.10)**	0.66 (0.13)	0.77 (0.15)	0.76 (0.18)	0.92 (0.09)	0.81 (0.17)	0.87 (0.11)
	KNN1	60	60	55	0.72 (0.10)	0.76 (0.09)	0.76 (0.11)	0.77 (0.16)	0.81 (0.18)	0.79 (0.20)	0.76 (0.24)	0.77 (0.22)	0.77 (0.24)
	LDA	60	60	50	0.75 (0.08)	0.75 (0.09)	0.78 (0.10)	0.68 (0.14)	0.75 (0.15)	0.75 (0.18)	0.85 (0.11)	0.81 (0.16)	0.83 (0.11)
	NB	15	35	45	0.71 (0.08)	0.69 (0.08)	0.75 (0.10)	0.66 (0.18)	0.84 (0.21)	0.79 (0.18)	0.92 (0.22)	0.86 (0.26)	0.84 (0.18)
	SVCL0	60	60	50	0.76 (0.08)	0.75 (0.08)	0.79 (0.10)	0.68 (0.13)	0.75 (0.15)	0.76 (0.18)	0.85 (0.13)	0.81 (0.15)	0.83 (0.12)
	SVCR0	60	50	35	0.76 (0.08)	0.76 (0.08)	0.78 (0.10)	0.82 (0.20)	0.79 (0.17)	0.84 (0.18)	0.96 (0.23)	0.80 (0.18)	0.84 (0.20)
				mean (sd)	*0.76 (0.24)*	*0.76 (0.26)*	*0.78 (0.23)*	*0.71 (0.39)*	*0.78 (0.39)*	*0.77 (0.36)*	*0.89 (0.40)*	*0.82 (0.39)*	*0.85 (0.39)*
Inj	CART0	50	60	60	0.76 (0.10)	0.75 (0.11)	0.79 (0.11)	0.72 (0.18)	0.78 (0.21)	0.80 (0.18)	0.88 (0.14)	0.87 (0.19)	0.82 (0.14)
	ENSMB	60	60	60	0.78 (0.10)	**0.75 (0.11)**	**0.78 (0.11)**	0.66 (0.18)	0.74 (0.20)	0.70 (0.17)	0.89 (0.12)	0.87 (0.18)	0.86 (0.13)
	KNN1	35	50	50	**0.68 (0.12)**	**0.72 (0.12)**	0.71 (0.12)	0.72 (0.21)	0.81 (0.23)	0.78 (0.19)	0.77 (0.25)	0.88 (0.28)	0.80 (0.28)
	LDA	55	60	60	0.74 (0.10)	0.73 (0.11)	0.74 (0.11)	0.71 (0.18)	0.73 (0.20)	0.71 (0.17)	0.82 (0.16)	0.87 (0.19)	0.81 (0.15)
	NB	25	10	45	0.69 (0.11)	0.69 (0.11)	0.74 (0.11)	0.64 (0.22)	0.81 (0.22)	0.68 (0.19)	0.90 (0.23)	0.88 (0.26)	0.84 (0.20)
	SVCL0	60	60	60	0.74 (0.10)	0.72 (0.11)	0.75 (0.11)	0.68 (0.19)	0.72 (0.19)	0.72 (0.16)	0.83 (0.18)	0.87 (0.18)	0.81 (0.16)
	SVCR0	60	60	60	0.76 (0.11)	0.75 (0.11)	0.76 (0.11)	0.71 (0.22)	0.70 (0.20)	0.79 (0.20)	0.93 (0.21)	0.87 (0.21)	0.92 (0.24)
				mean (sd)	*0.73 (0.23)*	*0.73 (0.25)*	*0.75 (0.22)*	*0.67 (0.37)*	*0.74 (0.37)*	*0.72 (0.34)*	*0.84 (0.39)*	*0.87 (0.41)*	*0.85 (0.39)*
Pro	CART0	40	55	60	0.73 (0.11)	0.76 (0.11)	0.77 (0.11)	0.67 (0.19)	0.77 (0.22)	0.78 (0.19)	0.88 (0.18)	0.75 (0.20)	0.80 (0.20)
	ENSMB	60	60	60	0.75 (0.10)	0.76 (0.11)	0.77 (0.11)	0.63 (0.18)	0.74 (0.23)	0.73 (0.19)	0.89 (0.14)	0.77 (0.21)	0.81 (0.19)
	KNN1	25	60	60	0.67 (0.12)	**0.71 (0.13)**	0.71 (0.13)	0.75 (0.22)	0.85 (0.26)	0.83 (0.25)	0.77 (0.31)	0.66 (0.31)	0.74 (0.30)
	LDA	50	60	60	**0.70 (0.11)**	**0.71 (0.12)**	**0.73 (0.11)**	0.67 (0.18)	0.72 (0.22)	0.73 (0.19)	0.82 (0.16)	0.73 (0.19)	0.80 (0.20)
	NB	15	60	55	0.66 (0.11)	0.69 (0.12)	0.70 (0.11)	0.66 (0.24)	0.80 (0.26)	0.74 (0.20)	0.91 (0.31)	0.75 (0.26)	0.81 (0.22)
	SVCL0	50	60	60	0.72 (0.11)	0.74 (0.12)	0.74 (0.12)	0.63 (0.17)	0.76 (0.23)	0.74 (0.19)	0.86 (0.16)	0.72 (0.21)	0.79 (0.21)
	SVCR0	60	60	60	0.74 (0.11)	0.76 (0.12)	0.74 (0.11)	0.68 (0.21)	0.71 (0.23)	0.74 (0.19)	0.94 (0.23)	0.86 (0.22)	0.89 (0.21)
				mean (sd)	*0.71 (0.24)*	*0.73 (0.24)*	*0.75 (0.25)*	*0.65 (0.39)*	*0.75 (0.38)*	*0.74 (0.39)*	*0.87 (0.40)*	*0.72 (0.36)*	*0.79 (0.40)*

Table S3.

Descriptor	Group	Technology	Target/Gene	Target/Family	Cell line
APR_CellLoss_72h_dn	A	Apredica	NA	cell cycle	HepG2
APR_MitoMass_72h_up	A	Apredica	NA	cell morphology	HepG2
APR_NuclearSize_72h_up	A	Apredica	NA	cell morphology	HepG2
ATG_NRF2_ARE_CIS	A	Attagene	NFE2L2	dna binding	HepG2
ATG_PPRE_CIS	A	Attagene	PPARD; PPARG; PPARA	nuclear receptor	HepG2
ATG_PXRE_CIS	A	Attagene	NR1I2	nuclear receptor	HepG2
ATG_VDRE_CIS	A	Attagene	NR1I1	nuclear receptor	HepG2
NVS_ADME_hCYP1A2	A	Novascreen	CYP1A2	cyp	cell-free
NVS_ADME_hCYP2C19	A	Novascreen	CYP2C19	cyp	cell-free
NVS_MP_rPBR	A	Novascreen	Tspo	transporter	cell-free
NVS_NR_hPXR	A	Novascreen	NR1I2	nuclear receptor	cell-free
OT_SRC1_SRC1FXR_1440	A	Odyssey Thera	FXR	nuclear receptor	HEK293T
ATG_BRE_CIS	B	Attagene	SMAD1	dna binding	HepG2
ATG_Oct_MLP_CIS	B	Attagene	POU2F1	dna binding	HepG2
NVS_ADME_hCYP2B6	B	Novascreen	CYP2B6	cyp	cell-free
NVS_MP_hPBR	B	Novascreen	TSPO	transporter	cell-free
NVS_TR_hNET	B	Novascreen	SLC6A2	transporter	NA
ATG_RARa_TRANS	C	Attagene	RARA	nuclear receptor	HepG2
NVS_GPCR_hOpiate_mu	C	Novascreen	OPRM1	gpcr	cell-free
NVS_NR_hAR	C	Novascreen	AR	nuclear receptor	cell-free
OT_AR_ARSRC1_0960	C	Odyssey Thera	AR	nuclear receptor	HEK293T
Tox21_AR_BLA_Antagonist_ratio	C	Tox21/NCGC	AR	nuclear receptor	HEK293T
Tox21_Aromatase_Inhibition	C	Tox21/NCGC	CYP19A1	cyp	MCF-7
Tox21_ERa_BLA_Antagonist_ratio	C	Tox21/NCGC	ESR1	nuclear receptor	HEK293T
Tox21_MitochondrialToxicity_ratio	C	Tox21/NCGC	NA	cell morphology	HepG2
Tox21_PPARg_BLA_Agonist_ch1	C	Tox21/NCGC	PPARG	nuclear receptor	HEK293T
Tox21_TR_LUC_GH3_Antagonist	C	Tox21/NCGC	THRB	nuclear receptor	GH3
APR_MitoticArrest_72h_up	D	Apredica	NA	cell cycle	HepG2
ATG_ERE_CIS_perc	D	Attagene	ESR1	nuclear receptor	HepG2
ATG_RXRb_TRANS	D	Attagene	RXRB	nuclear receptor	HepG2
NVS_NR_hER	D	Novascreen	ESR1	nuclear receptor	cell-free
NVS_NR_mERa	D	Novascreen	Esr1	nuclear receptor	NA
OT_ER_ERaERa_0480	D	Odyssey Thera	ESR1	nuclear receptor	HEK293T
Tox21_AR_BLA_Agonist_ch1	D	Tox21/NCGC	AR	nuclear receptor	HEK293T
Tox21_ERa_LUC_BG1_Agonist	D	Tox21/NCGC	ESR1	nuclear receptor	BG1
OT_NURR1_NURR1RXRa_0480		Odyssey Thera	RXRA	nuclear receptor	HEK293T

Table S4.

Descriptor	Group	Source	SMARTS	General structure
MACCS_141	A	MACCS	[CH3]	
MACCS_149	A	MACCS	[C;H3,H4]	
MACCS_160	A	MACCS	[C;H3,H4]	
MACCS_164	A	MACCS	[O,o]	
PUBCHEM_17	A	PUBCHEM	[#7]	
PUBCHEM_285	A	PUBCHEM	[#6]~[#7]	
PUBCHEM_351	A	PUBCHEM	[#6](~[#6])(~[#7])	
QPlogKp	A	QikProp		
PADEL_KR4080	B	PADEL	N	
PUBCHEM_340	B	PUBCHEM	[#6](~[#6])(~[#6])(~[#7])	
PUBCHEM_376	B	PUBCHEM	[#6](~[#7])(:[#6])	
PUBCHEM_416	B	PUBCHEM	[#6]=,:[#6]	
PUBCHEM_464	B	PUBCHEM	[#7][#6][#6]=,:[#6]	
PUBCHEM_524	B	PUBCHEM	[#6][#6]=,:[#6][#6]	
PUBCHEM_556	B	PUBCHEM	[#6]=,:[#6][#6][#6]	
PUBCHEM_640	B	PUBCHEM	[#6]=,:[#6][#6][#6][#6]	
PUBCHEM_660	B	PUBCHEM	[#6][#6]=,:[#6][#6][#6]	
MACCS_86	C	MACCS	[CH2][!C;!c;!#1][CH2]	
MACCS_102	C	MACCS	[!C;!c;!#1]~[O,o]	
MACCS_133	C	MACCS	*@*!:[N,n]	

PADEL_EStateFP54	C	PADEL	[ClD1]-*	
PADEL_KR2259	C	PADEL	[!#1]Cl	
PADEL_KR3821	C	PADEL	Cl	
PADEL_KR3869	C	PADEL	Clc1ccccc1	
PADEL_SubFP171	C	PADEL	[Cl][c]	
PUBCHEM_294	C	PUBCHEM	[#6]~[#17]	
PUBCHEM_342	C	PUBCHEM	[#6](~[#6])(~[#17])	
PUBCHEM_37	C	PUBCHEM	[#17]	
PUBCHEM_390	C	PUBCHEM	[#7](~[#6])(~[#6])	
PUBCHEM_501	C	PUBCHEM	[#17][#6]:[#6][#6]	
PUBCHEM_591	C	PUBCHEM	[#17][#6][#6][#6][#6]	
PUBCHEM_598	C	PUBCHEM	[#17][#6]:[#6]:[#6][#6]	
PUBCHEM_613	C	PUBCHEM	[#6][#7][#6][#6][#6]	
FP4_169	C	FP4	[Cl][c]	
QPlogBB	C	QikProp		
QPPMDCK	C	QikProp		
WPSA	C	QikProp		
CNS	C	QikProp		
MACCS_89	D	MACCS	[O,o]~*~*~*~[O,o]	
MACCS_91	D	MACCS	[!#6;!#1;!H0]~*~*~*~[CH2]~*	
MACCS_117	D	MACCS	[N,n]~*~[O,o]	

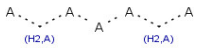
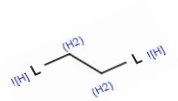
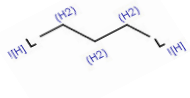
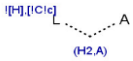
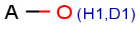
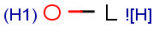
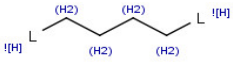
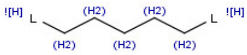
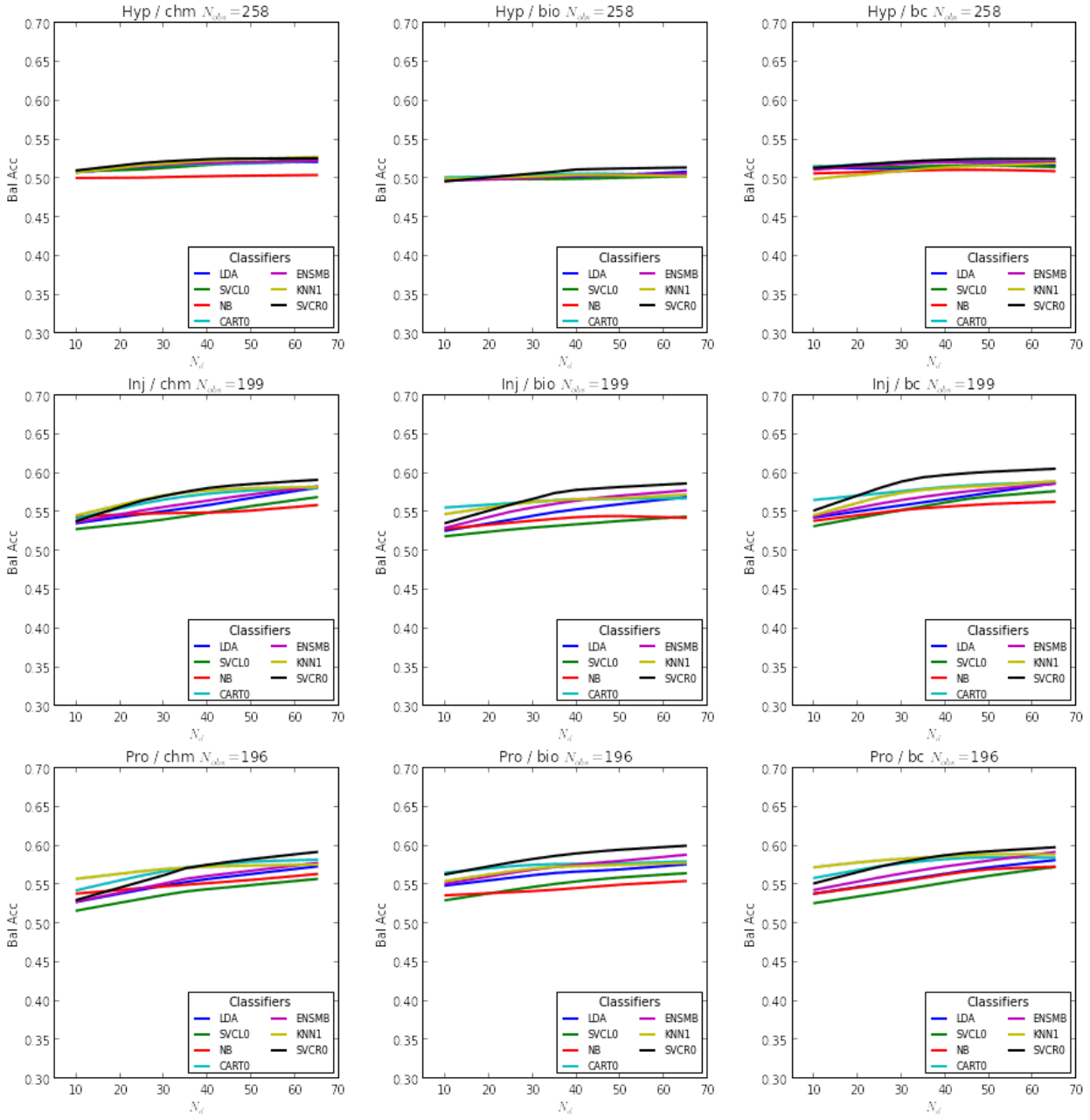
MACCS_128	D	MACCS	*~[CH2]~*~*~*~[CH2]~*	
MACCS_146	D	MACCS	[O,o]	
PADEL_KR341	D	PADEL	[!#1][CH2][CH2][!#1]	
PADEL_KR344	D	PADEL	[!#1][CH2][CH2][CH2][!#1]	
PADEL_SubFP2	D	PADEL	[CX4H2]([#6])[#6]	
FP4_2	D	FP4	[CX4H2]([#6])[#6]	
MACCS_138	E	MACCS	[!C;!c;!#1]~[CH2]~*	
PADEL_EStateFP34	E	PADEL	[OD1H]-*	
PADEL_KR1148	E	PADEL	[!#1][OH]	
PADEL_KR346	E	PADEL	[!#1][CH2][CH2][CH2][CH2][!#1]	
PADEL_KR347	E	PADEL	[!#1][CH2][CH2][CH2][CH2][CH2][!#1]	
FP3_27	E	FP4	[#6][OH]	
donorHB	E	QikProp		
FISA	E	QikProp		

Table S5.

		# desc.			BA			Sensitivity			Specificity		
Toxicity	Classifier	BIO	CHM	BC	BIO	CHM	BC	BIO	CHM	BC	BIO	CHM	BC
Hyp	CART0	45	55	35	0.51(0.09)	0.52(0.09)	0.53(0.10)	0.39(0.19)	0.42(0.15)	0.42(0.17)	0.63(0.18)	0.62(0.14)	0.63(0.15)
	ENSMB	50	45	45	0.51(0.10)	0.52(0.09)	0.53(0.10)	0.36(0.16)	0.4(0.16)	0.38(0.16)	0.65(0.15)	0.64(0.14)	0.67(0.14)
	KNN1	50	35	45	0.51(0.10)	0.53(0.09)	0.52(0.10)	0.53(0.14)	0.6(0.16)	0.55(0.17)	0.49(0.14)	0.45(0.16)	0.48(0.15)
	LDA	50	45	45	0.51(0.10)	0.52(0.10)	0.52(0.09)	0.43(0.15)	0.44(0.16)	0.45(0.15)	0.58(0.15)	0.6(0.14)	0.59(0.14)
	NB	40	65	50	0.51(0.07)	0.51(0.09)	0.51(0.08)	0.59(0.34)	0.56(0.26)	0.44(0.31)	0.43(0.34)	0.45(0.26)	0.59(0.30)
	SVCL0	50	35	45	0.50(0.10)	0.52(0.10)	0.52(0.09)	0.39(0.16)	0.43(0.17)	0.44(0.16)	0.62(0.16)	0.61(0.16)	0.6(0.15)
	SVCR0	50	35	55	0.51(0.09)	0.53(0.09)	0.53(0.10)	0.34(0.15)	0.39(0.16)	0.36(0.15)	0.69(0.13)	0.66(0.15)	0.7(0.13)
Inj	CART0	45	45	45	0.57(0.11)	0.58(0.11)	0.59(0.11)	0.33(0.20)	0.35(0.20)	0.37(0.20)	0.8(0.12)	0.81(0.11)	0.8(0.12)
	ENSMB	60	55	65	0.58(0.10)	0.58(0.11)	0.59(0.11)	0.3(0.19)	0.32(0.20)	0.33(0.19)	0.85(0.11)	0.84(0.10)	0.84(0.11)
	KNN1	65	45	55	0.58(0.12)	0.58(0.12)	0.59(0.12)	0.42(0.22)	0.45(0.22)	0.42(0.21)	0.73(0.13)	0.72(0.14)	0.76(0.13)
	LDA	60	60	65	0.57(0.11)	0.58(0.11)	0.59(0.11)	0.37(0.20)	0.37(0.21)	0.4(0.20)	0.77(0.13)	0.78(0.11)	0.77(0.12)
	NB	45	60	35	0.55(0.09)	0.56(0.11)	0.56(0.11)	0.71(0.30)	0.61(0.27)	0.5(0.30)	0.39(0.29)	0.5(0.23)	0.61(0.27)
	SVCL0	60	60	55	0.54(0.10)	0.57(0.11)	0.58(0.11)	0.23(0.17)	0.31(0.20)	0.33(0.20)	0.86(0.11)	0.82(0.11)	0.82(0.11)
	SVCR0	55	50	65	0.59(0.11)	0.59(0.11)	0.61(0.11)	0.32(0.19)	0.32(0.20)	0.35(0.19)	0.85(0.11)	0.86(0.10)	0.86(0.09)
Pro	CART0	35	40	40	0.58(0.10)	0.58(0.11)	0.59(0.11)	0.32(0.20)	0.35(0.21)	0.36(0.21)	0.84(0.11)	0.8(0.12)	0.81(0.11)
	ENSMB	60	65	60	0.59(0.10)	0.58(0.11)	0.59(0.11)	0.31(0.19)	0.33(0.20)	0.34(0.19)	0.87(0.09)	0.83(0.10)	0.84(0.10)
	KNN1	45	55	45	0.58(0.12)	0.58(0.12)	0.59(0.12)	0.44(0.22)	0.43(0.21)	0.42(0.21)	0.72(0.13)	0.73(0.13)	0.75(0.13)
	LDA	65	60	55	0.58(0.12)	0.57(0.12)	0.58(0.12)	0.38(0.21)	0.35(0.21)	0.37(0.21)	0.78(0.11)	0.79(0.12)	0.79(0.11)
	NB	40	55	45	0.55(0.09)	0.56(0.10)	0.57(0.10)	0.74(0.30)	0.73(0.23)	0.67(0.30)	0.35(0.29)	0.39(0.20)	0.47(0.26)
	SVCL0	45	60	55	0.56(0.10)	0.56(0.10)	0.57(0.11)	0.23(0.19)	0.28(0.20)	0.31(0.21)	0.89(0.08)	0.83(0.11)	0.83(0.11)
	SVCR0	50	55	60	0.6(0.10)	0.59(0.11)	0.6(0.10)	0.32(0.20)	0.31(0.20)	0.33(0.19)	0.87(0.09)	0.86(0.10)	0.86(0.09)

Figure S1.


Hyp / chm N_{obs} =258
Hyp / bio N_{obs} =258
Hyp / bc N_{obs} =258
Inj / chm N_{obs} =199
Inj / bio N_{obs} =199
Inj / bc N_{obs} =199
Pro / chm N_{obs} =196
Pro / bio N_{obs} =196
Pro / bc N_{obs} =196
Bal Acc
N_d
Classifiers
LDA
SVCL0
NB
CART0
ENSMB
KNN1
SVCR0